Formal dependency draft for the vanishing critical 2-variation prescribed-angle rectangle claim

Issue #134; companion specification for PR #122

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Abstract

This document is a formalization interface, not an unconditional Square Peg claim. It isolates the finite-partition and Liouville-primitive arguments that can be proved from the displayed definitions, states the external theorems with exactly the interfaces used, and records every unresolved dependency. In particular, no positive-size square and no four-distinct- vertices conclusion is asserted unless the nondegenerate conclusion of the Asano–Ike theorem is independently verified from the primary source.

1 Status vocabulary and theorem interface

Every lemma below carries one of the following statuses.

- `PROVED`: proved in this document from earlier displayed definitions and stated standard facts.
- `CITED`: imported from a named external source; citation fidelity remains part of the verification obligation.
- `STANDARD-CITED`: a standard theorem is used, but not reproved foundationally here.
- `STANDARD-ASSUMED`: a precise auxiliary interface is stated, but its formal proof and source pinning are outside this draft.
- `UNKNOWN`: the exact statement or a required construction has not been certified sufficiently to use as an assumable premise.

The intended dependency chain is

$$V_0 \longrightarrow \text{polygonal approximation} \longrightarrow \text{regular smoothing}$$
$$\longrightarrow \text{uniform local 2-variation} \longrightarrow \text{primitive compactness} \longrightarrow \text{Asano–Ike conclusion.}$$

The local consequences of V_0, the fourth box, and the elementary parts of the fifth box are proved below. The other boxes expose imported or unknown interfaces.

2 Vanishing 2-variation and polygonal error

Let $c\colon \mathbb{R} \to \mathbb{R}^2$ be continuous, 1-periodic, and injective on $[0, 1)$. For an interval $I = [a, b]$, define

$$\|x\|^2_{2\text{-var};I} := \sup_{a=t_0<\cdots<t_m=b} \sum_{j=0}^{m-1} |x(t_{j+1}) - x(t_j)|^2.$$

For a partition $D = \{0 = t_0 < \cdots < t_m = 1\}$ put

$$\operatorname{mesh}(D) = \max_j (t_{j+1} - t_j), \qquad w_2(c,\delta)^2 := \sup_{\operatorname{mesh}(D)\le\delta} \sum_j |c(t_{j+1}) - c(t_j)|^2.$$

Both suprema initially take values in $[0,\infty]$.

Definition 1 (Critical vanishing-variation hypothesis)**.** Write $V_0(c)$ for

$$\exists\delta_0 > 0 : \; w_2(c,\delta_0) < \infty \quad \text{and} \quad w_2(c,\delta) \longrightarrow 0 \quad (\delta \downarrow 0).$$

Thus every real-valued variation calculation below occurs in the finite regime $0 < \delta \le \delta_0$.

Lemma 2 (Lifted short-interval modulus)**.** *If $V_0(c)$ holds, $0 < \delta \le \frac{1}{2}$, and $I \subset \mathbb{R}$ has $|I| \le \delta$, then*

$$\|c\|_{2\text{-}var;I} \le \sqrt{2}\, w_2(c,\delta).$$

Status: PROVED.

Proof. If I does not cross an integer, extend any partition of I to a full-period partition of mesh at most δ. If it crosses one integer, insert the seam. The insertion replaces at most one increment $u+v$ by u, v, and $|u+v|^2 \le 2(|u|^2 + |v|^2)$. After reducing the two pieces modulo one, fill every complementary gap by subintervals of length at most δ and combine the nodes into a full-period partition. Since $|I| \le \delta \le 1/2$, at most one integer seam occurs. Take suprema. $\square$

For a partition D, let c^D be the affine interpolant of c on every interval of D.

Lemma 3 (Every-partition interpolation estimate)**.** *For every partition D of $[0,1]$,*

$$\|c - c^D\|_{2\text{-}var;[0,1]} \le \sqrt{8}\, w_2(c, \operatorname{mesh}(D)). \tag{2.1}$$

Consequently $V_0(c)$ implies $c^D \to c$ in 2-variation along every *sequence with* $\operatorname{mesh}(D) \to 0$.

Status: PROVED; replaces the load-bearing Yang quantifier.

Proof. Write $I_j = [t_j, t_{j+1}]$ and $y = c - c^D$. Since y vanishes at every t_j, inserting all partition nodes into an arbitrary variation partition costs at most a factor 2 in the squared sum. Indeed, although one crossing increment may meet many nodes of D, $y = 0$ at every inserted node, so only its two endpoint pieces can be nonzero. Hence

$$\|y\|^2_{2\text{-var}} \le 2 \sum_j \|y\|^2_{2\text{-var};I_j}.$$

On I_j, Minkowski's inequality and the fact that the affine chord has 2-variation equal to $|c(t_{j+1}) - c(t_j)|$ give

$$\|y\|_{2\text{-var};I_j} \le 2\|c\|_{2\text{-var};I_j}.$$

Near-optimal local partitions concatenate into one partition of mesh at most $\operatorname{mesh}(D)$, so

$$\sum_j \|c\|^2_{2\text{-var};I_j} \le w_2(c, \operatorname{mesh}(D))^2.$$

Combining the three inequalities proves (2.1). $\square$

External interface 4 (Simple interpolation, BG)**.** For every $\varepsilon > 0$ and every Jordan parametrization c, there is a partition D with $\mathrm{mesh}(D) < \varepsilon$ such that the parametrized affine interpolant c^D is a simple polygonal Jordan parametrization.

Status: CITED: Boedihardjo–Geng, Theorem 2.2; exact parameter-mesh and parametrized-interpolant wording must remain pinned.

Under Interface 4, choose D_n with mesh tending to zero and put $P_n = c^{D_n}$. The preceding proved lemma gives

$$\|P_n - c\|_{2\text{-var}} \to 0, \qquad P_n \to c \text{ uniformly}, \tag{2.2}$$

the uniform statement following because $P_n(0) = c(0)$.

3 The regular-smoothing interface

UNKNOWN obligation 5 (Regular parametrized corner smoothing)**.** For every parametrized simple polygon $P\colon \mathbb{R}/\mathbb{Z} \to \mathbb{R}^2$ and every $\varepsilon > 0$, construct a C^∞ *regular* Jordan embedding $Q\colon \mathbb{R}/\mathbb{Z} \to \mathbb{R}^2$ such that

$$\|Q - P\|_{1\text{-var}} < \varepsilon, \tag{3.1}$$

with compatible collars at every exit point and at the periodic seam. Here regular means that the chosen parametrization has nonzero velocity everywhere if that is part of the external smooth-Jordan convention.

Status: UNKNOWN at exact regularity; the geometric rounding is plausible, but the collar/jet/nonzero-speed parametrization has not been formalized.

Assuming Obligation 5, choose c_n from P_n with errors $\varepsilon_n \downarrow 0$. Since $\|x\|_{2\text{-var}} \le \|x\|_{1\text{-var}}$, (2.2) gives

$$\|c_n - c\|_{2\text{-var}} \to 0, \qquad c_n \to c \text{ uniformly}. \tag{3.2}$$

4 Uniform local variation

Define, for $0 < \delta \le \frac{1}{2}$,

$$\nu(\delta) = \sup_n \sup_{0 \le t-s \le \delta} \|c_n\|_{2\text{-var};[s,t]}, \tag{4.1}$$

where intervals are interpreted on the lifted periodic parameter.

Lemma 6 (Whole-family local modulus)**.** *Under (3.2), $\nu(\delta) \to 0$ as $\delta \downarrow 0$.*

Status: *PROVED, conditional only on the approximants supplied above.*

Proof. Put $e_n = \|c_n - c\|_{2\text{-var};[0,1]}$. Restriction, the seam insertion argument, and Lemma 2 give

$$\|c_n\|_{2\text{-var};[s,t]} \le \sqrt{2}\big(w_2(c, t-s) + e_n\big). \tag{4.2}$$

Given $\eta > 0$, first choose N so the right side is small for $n \ge N$, specifically $\sqrt{2}e_n < \eta/2$, then choose δ using $V_0(c)$ so $\sqrt{2}w_2(c,\delta) < \eta/2$. For the finite prefix $n < N$, each regular C^1 map is Lipschitz, say with constant L_n, and

$$\sum_i |\Delta c_n|^2 \le (\max_i |\Delta c_n|) \sum_i |\Delta c_n| \le L_n^2 \delta^2.$$

Taking the finite maximum proves the claim. ☐

5 Winding control and primitive compactness

Proposition 7 (Embedded-arc winding estimate)**.** *Assume Interface 8. Let $a\colon [u,v] \to \mathbb{R}^2$ be an injective rectifiable arc. Close it by the oriented chord from $a(v)$ to $a(u)$, obtaining L. Then*

$$\int_{\mathbb{R}^2} |\operatorname{Ind}(L,z)|\,dz \le \frac{\pi}{4}\|a\|^2_{2\text{-}var;[u,v]}. \tag{5.1}$$

Status: *PROVED conditional on the STANDARD-ASSUMED planar interfaces below.*

Proof. The components (α_j,β_j) on which the arc leaves its closing chord form disjoint excursions. Close each excursion by its subchord, forming a Jordan loop E_j with bounded domain Ω_j. The parameter intervals are disjoint, hence a single ordered partition gives

$$\sum_j \operatorname{diam}(E_j)^2 \le \|a\|^2_{2\text{-var};[u,v]}. \tag{5.2}$$

The planar isodiametric inequality gives $|\Omega_j| \le (\pi/4)\operatorname{diam}(E_j)^2$. As integral currents, the original loop is the mass-convergent signed sum of the excursion loops; the residual cycle supported on the chord line vanishes because one-forms restricted to a line have primitives. Planar filling uniqueness identifies the signed filling multiplicity with $\operatorname{Ind}(L,\cdot)$. The triangle inequality and (5.2) prove (5.1). □

External interface 8 (Excursion and planar-current package)**.** For an injective rectifiable arc and its closing chord: the off-chord parameter set is a countable union of open intervals whose endpoints land on the chord; every excursion plus its subchord is Jordan; its diameter is bounded by the diameter of the excursion together with its chord endpoints; and the associated signed boundaries are mass-summable and add to the original boundary. Moreover, compactly supported planar integral cycles have unique compactly supported two-dimensional fillings, whose integer multiplicity agrees almost everywhere with winding number. Finally, the planar isodiametric inequality is available.

Status: STANDARD-ASSUMED; each clause requires a source or formal proof.

Let

$$\lambda_0 = \frac{x\,dy - y\,dx}{2}, \qquad d\lambda_0 = dx \wedge dy,$$

and define normalized lifted primitives

$$F_n(0) = 0, \qquad F_n(t) - F_n(s) = \int_{[s,t]} c_n^*\lambda_0. \tag{5.3}$$

External interface 9 (Generalized Green formula)**.** For every closed rectifiable planar curve L,

$$\int_L \lambda_0 = \int_{\mathbb{R}^2} \operatorname{Ind}(L,z)\,dz. \tag{5.4}$$

Here L has its displayed orientation, $dx \wedge dy$ is the positive orientation of the plane, and Ind uses the corresponding counterclockwise-positive convention.

Status: CITED: Cufí–Verdera, main theorem at this regularity.

Proposition 10 (Primitive compactness)**.** *Assume Interfaces 8 and 9. The functions F_n have a subsequence converging locally uniformly on $\mathbb{R}$.*

Status: *PROVED CONDITIONAL from Lemma 6, Proposition 7, Interfaces 8 and 9, and STANDARD-CITED Arzelà–Ascoli.*

Proof. Uniform convergence places all images in one ball $B(0,R)$. For $0 \le t - s \le \delta$, close $c_n|_{[s,t]}$ by its chord. Equations (5.1)–(5.4) and direct chord parametrization yield

$$|F_n(t) - F_n(s)| \le \frac{\pi}{4}\nu(\delta)^2 + \frac{R}{2}\nu(\delta). \tag{5.5}$$

This is a common modulus tending to zero. A finite cover of $[0,1]$ bounds F_n uniformly there, so Arzelà–Ascoli supplies uniform convergence on $[0,1]$ along a subsequence. With $A_n = F_n(1)$,

$$F_n(k+r) = kA_n + F_n(r), \qquad k \in \mathbb{Z}, \quad 0 \le r < 1,$$

because the pulled-back integrand is periodic. This upgrades convergence to local uniform convergence on $\mathbb{R}$. □

For the alternative Liouville form $\lambda_{\mathrm{AI}} = y\,dx$,

$$y\,dx = -\lambda_0 + d(xy/2). \tag{5.6}$$

Thus its normalized primitive is

$$G_n(t) = -F_n(t) + \frac{x_n(t)y_n(t) - x_n(0)y_n(0)}{2}. \tag{5.7}$$

Equations (3.2), (5.6), and (5.7) prove local uniform convergence of G_n along the same subsequence.

Status: PROVED elementary exact-form conversion.

6 Conditional rectangle theorem

External interface 11 (Asano–Ike approximation criterion)**.** The exact primary theorem must be pinned to assert that the following data:

1. a Jordan parametrization c;
2. regular smooth Jordan parametrizations $c_n \to c$ parametrically in C^0; and
3. locally uniform convergence on $\mathbb{R}$ of normalized lifted primitives of the required Liouville form,

imply a prescribed-angle rectangle conclusion for every $\theta \in (0,\pi)$.

Status: UNKNOWN/CITED-UNVERIFIED interface; a pinned version, theorem/page, and faithful exact-hypothesis paraphrase must be independently checked.

UNKNOWN obligation 12 (Nondegeneracy and boundary order)**.** Verify from the primary theorem and its definitions that the θ-rectangle conclusion consists of four pairwise distinct points of $c(\mathbb{R}/\mathbb{Z})$, in the required cyclic boundary order, and is off the diagonal in the relevant configuration space. Also verify that θ is the angle between the diagonals and that $\theta = \pi/2$ therefore gives a positive-size classical square.

Status: UNKNOWN in this formalization; no nondegenerate square may be asserted from Interface 11 alone.

External interface 13 (Explicit nondegenerate conclusion AI_{ND})**.** For $\theta = \pi/2$, the conclusion imported from Asano–Ike supplies four pairwise distinct points of $c(\mathbb{R}/\mathbb{Z})$ in cyclic boundary order forming a nondegenerate Euclidean rectangle whose diagonals meet at angle $\pi/2$.

Status: UNKNOWN; this is the precise proposition needed from the source.

Lemma 14 (Perpendicular-diagonal rectangle)**.** *A nondegenerate Euclidean rectangle whose diagonals are perpendicular is a positive-size square.*

Status: *PROVED.*

Proof. For adjacent side vectors u, v with $u \perp v$, the diagonal vectors are $u+v$ and $u-v$. Their scalar product is $|u|^2 - |v|^2$; perpendicularity therefore gives $|u| = |v| > 0$. ☐

Theorem 15 (Analytic reduction to the cited rectangle criterion)**.** *Assume Interface 4, Obligation 5, Interfaces 8 and 9, and the unverified Interface 11. If c is a Jordan parametrization with $V_0(c)$, then c satisfies the approximation and primitive-convergence hypotheses listed in Interface 11. Consequently c has* exactly the conclusion stated by the verified version *of the cited Asano–Ike theorem.*

Status: *PROVED CONDITIONAL REDUCTION; not a nondegenerate rectangle claim.*

Proof. Interface 4 and the proved interpolation estimate produce the simple polygons P_n. Obligation 5 produces regular smooth Jordan c_n satisfying (3.2). Lemma 6 gives the common local variation modulus. Proposition 10, Interface 9, and (5.6)–(5.7) give the locally uniformly convergent lifted primitives. These are precisely the data listed in Interface 11; therefore only the verified conclusion of that interface may be imported. ☐

Corollary 16 (Positive square, conditional on nondegeneracy)**.** *Under all hypotheses of the preceding theorem and assuming the explicit Interface 13, c inscribes a positive-size square with four distinct vertices in cyclic boundary order.*

Status: *CONDITIONAL; not available while Interface 13 is UNKNOWN.*

7 Machine-facing dependency ledger

Component	Status	Exact interface used
Fine-mesh/local modulus	PROVED	Definition V_0; seam factor $\sqrt{2}$
PL interpolation error	PROVED	Every partition; constant $\sqrt{8}$
Simple PL Jordan choice	CITED	BG theorem, parameter-mesh version
Regular smoothing	UNKNOWN	regular C^1/C^∞ embedding; (3.1)
Whole-family modulus	PROVED	tail-first then finite-prefix quantifiers
Arc winding estimate	CONDITIONAL	excursion/current package
Generalized Green	CITED	arbitrary closed rectifiable planar curve
Primitive compactness	PROVED CONDITIONAL	current/Green interfaces, common modulus, Arzelà–Ascoli
Liouville conversion	PROVED	equations (5.6)–(5.7)
AI approximation criterion	UNKNOWN/CITED-UNVERIFIED	version and exact interface must be pinned
Four distinct vertices	UNKNOWN	off-diagonal/noncollapse conclusion
$\theta = \pi/2$ square	CONDITIONAL	follows only after previous row

8 Source identifiers (pinning incomplete)

The imported interfaces have the following source candidates. A version and PDF-page reference is part of a complete pin; any missing page below is therefore an explicit verification blocker rather than an immutable pin.

- Boedihardjo–Geng: arXiv:1309.1576v2, Theorem 2.2, PDF pp. 6–7, `https://arxiv.org/abs/1309.1576v2`.
- Cufí–Verdera: arXiv:1306.6832v1, main theorem, PDF p. 1 (specialized to the relevant linear test form), `https://arxiv.org/abs/1306.6832v1`.
- Asano–Ike: arXiv:2412.21057v3, Theorem 1.1, PDF p. 2; nondegeneracy discussion at PDF pp. 2–3 and the off-diagonal argument in the proof of Theorem 4.1 at PDF p. 19 remain subject to independent review, `https://arxiv.org/abs/2412.21057v3`.

9 Scope

This draft neither proves the full Square Peg conjecture nor promotes PR #122 beyond `sketch`. It does not use PRs #115, #118, or #119 as premises. There is currently no problem-specific `problems/square-peg/RULES.md`; consequently the root repository rules are the only status protocol available, and this missing problem-level protocol is itself a governance gap rather than permission to promote.